

Final Spring 26 AST1002

- 1) extra points: _____
The first pulsar was detected with a
A) X-rays telescope B) Hubble telescope
C) radio telescope D) IR telescope
- 2) _____
A) red dwarf
B) yellow dwarf
C) blue stars
D) brown dwarf.
E) Which types of stars last "for ever" because they burn their fuel slowly (slow metabolism. last forever).
- 3) extra points: _____
Jocelyn Bell named the signals coming from the first pulsar
A) Little Green Men B) Big Green Men
C) Strange Green Men D) Pulsating Green Signal
- 4) What is the upper limit to the mass of a white dwarf? _____
A) about 2.8 solar masses
B) about 1.4 solar masses
C) There is no upper limit.
D) about 1 solar mass
E) There is an upper limit, but we do not yet know what it is.
- 5) What is a white dwarf? _____
A) the exposed core of a dead star, supported by neutron degeneracy pressure
B) a cool and very small main-sequence star with a mass of less than 1.4 solar masses
C) a hot, but very small main-sequence star with a mass of less than 1.4 solar masses
D) the exposed core of a dead star, supported by electron degeneracy pressure
E) the name for the singularity at the center of a black hole
- 6) The classification of stars established by Annie Jump Cannon is _____
A) A,O,B, G,F,K,M,L B) M,B,A, F,G,K,O,L
C) A,B,O, F,G,K,M,L D) O,B,A, F,G,K,M,L
- 7) When galaxies are moving away from us, they are _____
A) redshifted B) greenshifted
C) There is no shift D) blueshifted

- 8) What produces a type-I supernova? 8) _____
 A) the radioactive decay of nickel 56 into cobalt 56 into iron 56
 B) a nova igniting a helium flash in its red giant companion
 C) mass transfer onto a white dwarf pushing it over 1.4 solar masses
 D) the helium flash blows apart a giant's core
 E) the collapse of the core of a massive star
- 9) After a massive-star supernova, what is left behind? (hint: not a supernova type IA but type II) 9) _____
 A) always a neutron star
 B) always a white dwarf
 C) always a black hole
 D) either a neutron star or a black hole
 E) either a white dwarf or a neutron star
- 10) When a star is on the main sequence, the luminosity depends mainly on the 10) _____
 A) does not depend on temperature
 B) temperature to the 4th power
 C) temperature
 D) temperature to the 2nd power
- 11) In our Sun, the last fusion stage will produce 11) _____
 A) He
 B) C, O
 C) Fe
 D) Mg
- 12) 2 neutron stars orbiting each other can produce (shaking the fabric of space-time) 12) _____
 A) sound waves
 B) burst of neutrinos
 C) gravitational waves.
 D) a red dwarf
- 13) What do we call a "failed star" ? (can only burn deuterium to lithium). we don't want to hurt 13) _____
 feeling)
 A) brown dwarf
 B) red dwarf
 C) neutron star
 D) yellow dwarf
- 14) The star that burns twice as bright burns ... (hint: blade runners) 14) _____
 A) half as long
 B) one fourth as long
 C) one third as long
 D) just as long
- 15) We studied the distance ladder. Which technique is at the top of the ladder. To measure the 15) _____
 distance of extremely far objects? (what is extremely luminous and powerful)
 A) supernovae type IA
 B) parallax method
 C) shift in luminosity in HR diagrams
 D) cepheid variables
 E) echolocation with microwaves
- 16) What is a planetary nebula? (hint: think Sun) 16) _____
 A) the ejected envelope, often bipolar, of a red giant surrounding a stellar core remnant
 B) a planet surrounded by a glowing shell of gas
 C) the bipolar jets ejected by a T Tauri variable
 D) a type of young, medium mass star
 E) the disc of gas and dust surrounding a young star that will soon form a solar system

- 17) During their main life, stars burn hydrogen to _____
A) lithium
B) helium
C) Stars use coal as their fuel
D) deuterium
E) magentium
- 18) the sun peaks in the _____
A) red B) green C) blue D) infra red
- 19) the Algol system is called the _____
A) demon star B) dog star C) king star D) whale star
- 20) A star (no matter what its mass) spends most of its life: _____
A) as a planetary nebula.
B) as a main sequence star.
C) as a protostar.
D) as a T Tauri variable star.
E) as a red giant or supergiant.
- 21) When a star's inward gravity and outward (gas) pressure are balanced, the star is said to be _____
A) in hydrostatic equilibrium.
B) in thermal expansion.
C) in rotational equilibrium.
D) a stage 2 protostar.
E) in gravitational collapse.
- 22) whats the most important stellar parameter of stars? _____
A) composition B) luminosity C) rate of spin D) mass